

DengueRadar

Team meeting pack: we are changing focus. This is the new problem, solution, plan and who does what.

When	28 Sep 2026, 10pm SGT (about 60 minutes)
Who	Braedon (lead, platform, model) Jingyi (weather data) Ziqi (dengue and places data, then demo) Komal (story and pitch)
Goal	Agree the new direction, leave knowing your tasks and due dates, and decide the open questions
Hard deadline	Round 1 PDF (3 slides) on Devpost by 6 Oct, 11:59pm SGT . Our target is 4 Oct
Next checkpoint	1 Oct: go or adjust, based on whether the archived data supports a per-area forecast

The new line. NEA's public dashboard shows today's clusters. DengueRadar shows where risk is heading in the next two weeks, why, and what to do.

What changed and why

We had built the plan around senior vulnerability and Active Ageing Centre staff. The official brief is different: a forward-looking dengue risk forecast **by planning area**, a near-real-time pipeline with streaming or incremental ingestion, and a predictive model on an interactive map, aimed at Singapore's communities. We are realigning to it and adding a **prescriptive** layer: what to do about each forecast.

	Before (27 to 28 Sep)	Now (from tonight)
Problem	Seniors are least likely to be tested in time	NEA's public dashboard is reactive: clusters appear after people are sick
User	Active Ageing Centre staff	Singapore residents, and community leaders (town councils, RCs, condo MCSTs). NEA is a partner
Forecast	National model trained; per-area is a hand-set score	Trained 2-week risk forecast for each planning area, plus national context
Output	Visit-first block list for centres	Island view and Area view: risk, top drivers, and an action plan with owner and timing
Seniors	The headline	A secondary 'extra care' line (keeps Track A fit)
Ingestion	Daily cluster archive	Daily cluster archive plus incremental weather loading (brief asks for it)

Full record: `_bmad-output/planning-artifacts/sprint-change-proposal-2026-09-28.md`. All repo docs were updated to match; the four earlier forges are kept as history.

SECTION 1

Agenda and decisions for tonight

Min	Item	Lead	Outcome we need
0-8	Why we are changing: the brief vs our plan (cover page); the new idea in one minute	Braedon	Everyone can say the new idea back
8-18	Status round: done / blocked / next (2 min each)	All	Blockers named
18-38	Decisions D1 to D7 below	Braedon	Yes/no and an owner for each
38-48	Data: what changes for Jingyi and Ziqi (training panel, station map)	Jingyi, Ziqi	Tasks and 1 Oct dates agreed
48-55	The 3 slides: what Komal needs, and what we may and may not claim	Komal	Draft dates locked
55-60	Read-out of next 3 tasks per person; pick weekly check-in	All	No surprises

Decisions we need tonight

	Decision	Recommendation	Owner
D1	Approve the pivot to planning-area forecasting for residents and community leaders.	Yes. It matches the official brief and gives a sharper gap against NEA's public dashboard.	All
D2	Train the per-area forecast on the cited cluster archives (SGCharts 2015 to 2020, Nature 2020), or keep a hand-set score?	Train it, keep the points score as a baseline and fallback. Decide for good at the 1 Oct checkpoint.	Braedon, Ziqi
D3	Keep seniors as a secondary 'extra care' flag? Track A is Caring for an Ageing Singapore.	Yes: cheap (Census data is already loaded) and protects problem fit (30% of score).	All
D4	How to word the NEA gap. NEA has a national forecast and a yearly 1 km2 risk map.	Always say 'NEA's public cluster dashboard'. Komal confirms wording from NEA's pages by 30 Sep.	Komal
D5	Prescriptive actions. Who reviews the action wording?	Braedon drafts, Komal checks against NEA's public prevention advice; try to reach one town council or RC contact.	Braedon, Komal
D6	Team name and Devpost registration; collect member details.	Choose tonight; Komal registers.	All, Komal
D7	Shared files and check-in time.	Code and docs in GitHub, live tables in Databricks. Pick a fixed 30-minute slot.	All

SECTION 2

The problem and why it matters

Dengue is hyperendemic in Singapore and outbreaks affect tens of thousands of residents. Rising temperatures and erratic rainfall extend the Aedes breeding season. NEA's public cluster dashboard tells us where clusters already are, when people are sick and mosquitoes have been breeding for weeks. Nothing public tells residents and community leaders where risk is heading in the next two weeks, or what to do.

Fact	Number	Source	Status
Dengue cases, 2020 (record year)	35,261 (DATASETS.md) or 35,012 (old checkpoint doc)	MOH weekly bulletin	CONFIRM: Ziqi picks one
Live NEA clusters, 28 Sep	11 clusters, 117 cases; largest Taman Jurong (80)	NEA Dengue Clusters	Checked
How NEA's public dashboard works and how a cluster is defined	Clusters only, updated daily	NEA website	CONFIRM: Komal
Warmer, wetter weather extends the Aedes season	Not sourced yet	NEA or a study	CONFIRM: Komal
Extra-care line: residents 65+	614,380 (15.2%); 88,400 65+ live alone (national)	SingStat	Checked

SECTION 3

Our solution

DengueRadar forecasts dengue risk for every planning area two weeks ahead, explains the drivers, and tells residents and community leaders what to do, so action starts before clusters form.

Step	What it does
1 Ingest	Daily NEA cluster archive (running since 28 Sep) and incremental loading of 5-minute weather readings into Delta
2 Engineer	Weather lags, hot days, neighbouring-area clusters, recurrence, block age, all keyed by planning area and week
3 Forecast	Trained model (LightGBM in MLflow) predicts cluster activity per planning area for the next 2 weeks (Low / Medium / High); SHAP shows the top 3 drivers
4 Prescribe	The leading driver maps to actions with an owner and a timing: what residents do, what town councils, RCs and MCSTs inspect or send
5 Show	Island view (map of 55 planning areas), Area view (why and what to do), Genie for plain-English questions, resident alert mock

User	Decision it changes
Residents	What to check at home this week; when to see a doctor early
Town councils, RCs, condo MCSTs	Where to inspect drains and common areas first; where to send reminders
NEA (partner)	Public-area causes (drains, hawker centres) route to them

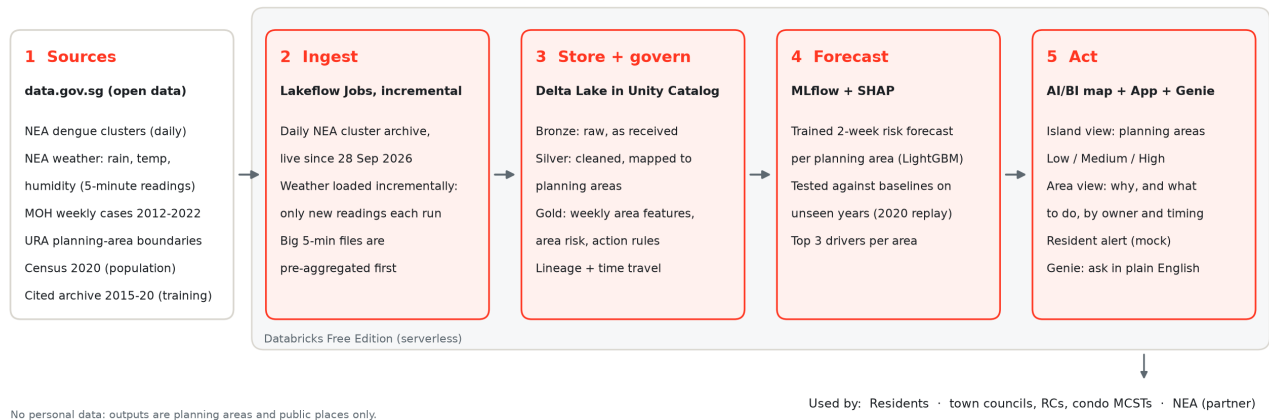
Be exact about the target. The archive only holds cases inside NEA clusters. We predict **cluster activity** in a planning area over the next 2 weeks, not total dengue cases. Say it on the slides.

SECTION 4

Architecture on Databricks, and what is proven

DengueRadar on Databricks Free Edition

From reactive dashboards to a 2-week forecast with a plan of action. Components tested on our Free Edition workspace on 28 Sep 2026.



Free Edition check, 28 Sep 2026

Area	Test	Result
Egress	Notebook reaches data.gov.sg: download, datastore API, real-time rainfall API	Yes (4/4)
Unity Catalog + Delta	Own schema and tables; time travel	Yes
Lakeflow Jobs	Daily 21:00 SGT NEA cluster snapshot	Yes. Live since 28 Sep (day 1: 11 clusters, 117 cases)
Spatial SQL	st_dwithin, H3, shapely fallback	Yes
MLflow	Log a run; register a model in Unity Catalog	Yes
AI/BI dashboard	Point map of today's clusters (draft Island view)	Yes
Genie	'Which cluster has the most cases?'	Yes: Taman Jurong, 80
Databricks Apps	Streamlit template starts	Yes
Incremental ingestion	Structured Streaming to Delta; Auto Loader picks up only new files; MERGE watermark fallback	Not run yet. Added 28 Sep (notebook 00, section 5). Braedon runs by 29 Sep

'Real-time' has to be described honestly: NEA clusters update daily; weather is every 5 minutes; the forecast refreshes daily. Free Edition limits: serverless, one 2X-Small SQL warehouse, up to three Apps, quota-limited compute. Pre-aggregate the big weather files before upload.

SECTION 5

Data: what we use, what changed, the gaps

Dataset	Feeds	Tier now	Owner
NEA Dengue Clusters (daily snapshot)	Live input; cluster cases per area; action tags	Core	Ziqi
SGCharts archive 2015 to 2020; Nature Scientific Data 2022 (CC BY 4.0), cited	Training panel for the per-area forecast; 2020 replay	Core (training) (was validation only)	Ziqi
NEA rainfall, temperature, humidity: historical and real-time	Weather features; incremental ingestion demo	Core (real-time was Should)	Jingyi
Weather station locations	Map each station to a planning area	New, Core	Jingyi
MOH weekly cases 2012 to 2022	National context feature; Level 1 forecast	Core	Ziqi
URA planning-area and subzone boundaries	Forecast unit (55 areas); maps	Core	Ziqi
Census 2020 residents by age	Population denominator; 65+ extra-care flag	Should (was Core)	Ziqi
NEA high-Aedes zones (2019 survey)	Candidate feature; weak, static	Core, kept if it helps	Ziqi
Parks; hawker centres	Candidate features; places-where-people-gather actions	Should	Jingyi
HDB property info and buildings	Block age (H3); optional block detail	Should	Ziqi
CHAS clinics	Nearest subsidised test in resident action	Should (was Could)	Ziqi

Gaps and what we do

Gap	Plan
Archive holds only cases inside NEA clusters	Target is cluster activity, said openly on slides
2020 is dense to May, then one snapshot a month; Nature covers Feb to Jul 2020	Use Nature for the surge; report usable weeks and areas; this is the 1 Oct go/adjust test
Training panel is small (55 areas, about 270 weeks; about half the areas residential)	Honest error bars; time-ordered validation; leave-areas-out check; points score as fallback
MOH weekly cases end Dec 2022	Find Communicable Diseases Agency bulletins; format to check
High-Aedes zones come from a 2019 survey	Keep only if validation improves; label as 2019
Seniors living alone by area: not available (national 88,400)	Use 65+ share for the extra-care flag
No official per-area case history	Why the archive matters; cite it

SECTION 6

Model: hypotheses, validation, metrics

Hypotheses (datathon handbook method)

ID	Hypothesis	Features
H1 Weather lag	Warm periods, and rain 2 to 4 weeks earlier followed by dry spells, raise next-fortnight cluster activity. Our monthly chart supports temperature (+0.29 at 4 months), not rain alone	hot_days_31c, rain_lag2..4, temp_lag, humidity_lag
H2 Spatial spillover	Clusters in neighbouring areas raise an area's own risk	cluster_cases_adjacent, dist_to_nearest_cluster_km
H3 Recurrence	Areas with clusters in the past year recur, especially with older blocks	cluster_weeks_last_52, median_block_age

Any of these may be disproven. Saying so honestly is part of the pitch.

What we train, and what we beat

Item	Definition
Target	Cluster cases in an area over the next 2 weeks; 'Rising' = at least 20% above its 4-week average (with a minimum count)
B1 Persistence (main)	Next 2 weeks = the last 4 weeks' average ('clusters carry on')
B2 Seasonal	Same week averaged over training years
B3 Points score	The old hand-set score (cluster size, Aedes zone, national Rising, rain, hot days)
Model	LightGBM regression and Rising classifier in MLflow; Poisson or logistic as a readable cross-check; SHAP for drivers

Validation: how we test honestly

- **Time-ordered only.** Train 2015 to 2018, test 2019; train to 2019, test Jan to Jul 2020. Never a random split.
- **Leave-areas-out check** (GroupKFold by planning area): the model must learn patterns, not memorise areas.
- **No leakage.** Features for week t use only data available before week t; NEA publishes about a week late.
- **Known weakness.** Published Singapore models catch timing but underpredict the size of peaks. So will ours.

Metrics and the cost matrix

Metric	Meaning	Target
Rising recall / precision, PR-AUC	Of real rises, how many we flag; of flags, how many were real	Recall first; above B1 and B3
Hit rate	Share of new or growing clusters that were in areas rated High	Above B1; compare NEA's ~90% for its 1 km2 map
Lead time (weeks)	Weeks before an area's cluster cases crossed the alert size that we first flagged it	Measure it. Never claim it before
Alert load	Share of areas rated High in a typical week	At most 15%
MAE / MAPE	Average miss (cases per area; national cases per week)	Below baselines; report beside NEA's 17% to 24%

Threshold from the cost matrix: a missed rising area costs 5 times an unnecessary reminder (a starting assumption, labelled as one). Choose the threshold that minimises expected cost on the validation years, then check alert load stays at most 15%.

SECTION 7

The prescriptive layer: what to do about each forecast

Prediction without action is another dashboard. Each area's top SHAP driver maps to actions. **Wording is a draft: check against NEA's public prevention advice before any slide uses it.**

Leading driver	Resident action (this week)	Community leader action	Timing
Rain then warm spell (H1)	Empty flower-pot plates and containers, clear gully traps, cover storage	Town council: clear drains and roof gutters. MCST: check common-area water	Within 7 days
Active cluster nearby (H2)	Repellent, long sleeves at dawn and dusk, see a doctor early for fever (nearest CHAS clinic)	RC and town council: door-to-door reminders around the cluster	Within 3 days
Neighbouring area rising (H2)	Heightened awareness; do the weekly checks	Coordinate with the neighbouring council; pre-position inspection	Before next week
Recurrence, older blocks (H3)	Routine weekly checks	Town council: scheduled inspection of known recurring spots	Within 14 days
Places people gather (if validation keeps them)	Repellent before outdoor meals or exercise	NEA and NParks partners check drains and bins	Within 7 days

- **Extra-care flag:** where an area's 65+ share is in the top third, add 'check on older neighbours and family members'. This is the only place seniors enter the product.
- **What-if sliders** are illustrative only. We have no evidence for effect sizes, so we never show a percentage reduction as fact.
- **Outputs:** Island view (55 planning areas, Low / Medium / High), Area view (risk, drivers, actions), resident alert mock (phone message, never sent), 2020 replay.

Example resident alert (mock)

Dengue risk in Bedok is forecast HIGH for the next 2 weeks (warm spell, a cluster nearby). This week: empty plates and containers, use repellent, see a doctor early for fever.

Why judges should like it

- **Originality (30%):** predicts, explains, and prescribes; not a generic map.
- **Feasibility (25%):** named datasets, Free Edition components tested, live archive running, honest limits.
- **Impact (30%):** a two-week window is when clearing water and inspecting drains works best; measured by lead time and hit rate.

SECTION 8

Where we stand, and the risks

Done

- Workspace created; every Free Edition component passed; daily NEA cluster archive running since 28 Sep.
- Ziqi: 17-dataset table, archive inspection (SGCharts 256 clean snapshots; Nature 213 of 213 subzones mapped), weekly dengue CSV, boundaries, slide-2 lines.
- Braedon: architecture diagram (updated tonight), model definitions (rewritten tonight), monthly lag chart, all repo docs updated to the new direction.

Not done yet

Task	Owner	Due
Run the incremental-ingestion check (streaming, Auto Loader, MERGE)	Braedon	29 Sep
Weekly rainfall and temperature CSV; station-to-planning-area map	Jingyi	29 Sep / 1 Oct
Per-area weekly training panel and coverage table	Ziqi	1 Oct
Cited statistics, NEA-dashboard description, resident story, mock-ups	Komal	30 Sep to 1 Oct
Weekly lag chart; prescriptive rule table draft	Braedon	30 Sep to 1 Oct

Risks

Risk	What we do
Thin or gappy per-area training data could give a weak model	1 Oct go/adjust test; fallback to points score; report limits honestly
Target is cluster activity, not total cases	Name it on slide 2; do not say 'predicts dengue cases'
'NEA does not predict' is too broad (NEA has a national forecast and a yearly risk map)	Say 'public cluster dashboard'; acknowledge NEA's models on the slide
Streaming untested on Free Edition	Run check; fallback to scheduled incremental MERGE job; do not call daily clusters 'live streaming'
Lead time unmeasured	Measure on the backtest; reword the pitch to what the data shows
Action wording unsupported	Check against NEA advice; no effect-size claims; try one town council or RC contact
Track fit (Caring for an Ageing Singapore) weakens without seniors	Keep the extra-care flag; confirm track wording on Devpost
Free Edition quota lock during a demo	Small data; rehearse in the demo workspace; 30-second backup video by 25 Oct

SECTION 9

Who does what, by when

Person	Task	Due	Done when
Braedon Lead, platform, model	Run incremental-ingestion check; post three yes/no lines	29 Sep	Results in team doc
	Weekly lag chart, shifts 0 to 8 weeks, temperature / rain / humidity (tests H1)	30 Sep	One chart shows the best lag
	Prescriptive rule table draft with owner, timing and source link	1 Oct	Komal can review it
	Hand diagram and chart to Komal as PNG; run the 1 Oct checkpoint; judges-panel mock	1 to 3 Oct	Go/adjust decision recorded; top 5 judge questions
Jingyi Weather data	Weekly national rainfall and temperature CSV	29 Sep	Braedon can chart it
	Weather rows in DATASETS.md; real-time API samples; station locations	30 Sep	Link, format, range, granularity each
	Map each station to a planning area; list areas with no nearby station	1 Oct	CSV plus gap note
	Weather line for slide 2 to Komal	1 Oct	Pasted in team doc
Ziqi Dengue and places, then demo	Weekly per-planning-area training panel from SGCharts and Nature; coverage table; can June to Dec 2020 be used?	1 Oct	CSV (area_id, week_start, cluster_cases) plus coverage
	Reconcile the 2020 case count; find where post-2022 weekly cases are published	1 Oct	One number with source; format noted
	Dengue and population lines for slide 2; boundary image for mock-up	1 Oct	Komal has them
	Explore dashboard, Genie and Apps by hand for the map	11 Oct	Recommendation with reasons
Komal Story and pitch	Register on Devpost; pick team name; collect details; confirm problem code and track wording	Tonight	All fields for every member
	Cited statistics: outbreak sizes, NEA dashboard and cluster definition, weather-and-Aedes evidence, cost of dengue	30 Sep	Number, link, year each
	Resident story (3 sentences); Island view, Area view and resident-alert mock-ups (labelled illustrative)	30 Sep to 1 Oct	Images ready
	Draft 3 slides; fix after mock judging; export PDF; submit	2, 3, 4 Oct	Devpost confirmation screenshot
Everyone	Read the change proposal; send Komal your details; post blockers same day; review final PDF	28 Sep to 4 Oct	No surprises on 4 Oct

SECTION 10

Timeline, slides, judging and team rules

Date	Milestone
28 Sep (tonight)	This meeting. Decisions D1 to D7.
28 to 30 Sep	Data verification; training panel; statistics and story; incremental check; lag chart
1 Oct	Checkpoint: does the archive give a usable per-area signal? Go on the trained model, or fall back
2 to 3 Oct	Komal drafts 3 slides; 3 Oct mock judging with the hardest questions
4 Oct	Final polish, export PDF, submit on Devpost (our target)
5 to 6 Oct	Buffer. Hard deadline 6 Oct, 11:59pm SGT. Top 10 announced 9 Oct
12 to 26 Oct	Mentored sprint: 15 Oct walking skeleton; 19 Oct all MUST items working; 25 Oct feature freeze and backup video; 26 Oct rehearse 5 times
27 Oct, 4 to 7pm	Demo Day. Finale 4 Nov

The 3 slides

Slide	Draft answer
1 Problem	NEA's public dengue dashboard shows clusters after people are already sick, so residents and community leaders find out too late to prevent them. Number: outbreak sizes (MOH) and today's live clusters. Why now: rising temperatures and erratic rain extend the Aedes season
2 Solution and data	DengueRadar forecasts risk for every planning area two weeks ahead, explains the drivers, and tells residents and community leaders what to do. Datasets: NEA clusters, NEA weather, MOH cases, URA boundaries, Census; training data cited (SGCharts, Nature). Different from a dashboard: predicts, explains, prescribes
3 Architecture and impact	Lakeflow Jobs and incremental loading, Delta bronze/silver/gold in Unity Catalog, MLflow with SHAP, AI/BI map or App, Genie. Demo: Island view, Area view, 2020 replay. Impact: lead time, hit rate vs baseline, alert load

Do not put on the slides

- 'Two weeks early' as a proven fact: lead time is unmeasured.
- 'NEA does not predict': say 'NEA's public cluster dashboard is reactive'.
- 'More rain means more dengue': our own chart does not support it.
- That we predict total dengue cases: we predict cluster activity.
- Any effect size for an action ('cleaning drains cuts risk by X%').
- 'Parks or hawker centres predict dengue': say 'places where people gather'.
- More archive days than we have: say 'running since 28 Sep'.

Round 1 judging

Criterion	Weight
Problem fit and social impact	30%
Solution quality and originality	30%
Data and technical feasibility	25%
Clarity of submission	15%

Team rules

- One owner per notebook or folder; .py source; branch, merge to main only when it runs.
- Keep data small on Free Edition; stop idle notebooks, jobs and apps.
- No personal data or credentials anywhere; the resident alert is a mock-up.
- Credit every dataset, library and AI tool. Post blockers the same day.